
Dynamic Curriculum Ordering with Dual Signals for Continual Learning

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Abstract

1 Catastrophic forgetting in continual learning is notoriously sensitive to task order,
2 especially for large transformers where an unlucky permutation can waste days of
3 compute. We address the open problem of online task ordering with DyCLOS, a
4 model-agnostic curriculum that schedules tasks on the fly using two lightweight
5 signals computed from a small probe per task. The first is data similarity measured
6 via an optimal-transport dataset distance in a shared frozen embedding space, and
7 the second is gradient interference estimated as the cosine of adapter gradients.
8 DyCLOS constructs a dynamic curriculum graph with edge costs that combine
9 dissimilarity and interference, then maintains a minimum-cost Hamiltonian path
10 prefix by solving a sliding-window beam search as tasks arrive. Training updates
11 are limited to parameter-efficient APD-style adapters, and a stability-aware inner
12 loop attenuates transient performance drops. We implement a reproducibility
13 toolkit and evaluate DyCLOS in three settings: a synthetic-vision sanity check, a
14 compact CIFAR class-incremental stream, and a synthetic NLP multi-task stream.
15 Across these, DyCLOS improves final accuracy by 0.9-3.3 percentage points over
16 fixed or random orders and reduces average forgetting by 33-88%, with small
17 absolute overhead; ablations verify that combining both signals yields the best
18 stability-plasticity trade-off. These findings show task ordering is a controllable
19 lever and that DyCLOS offers a practical, modality-agnostic curriculum layer for
20 continual training pipelines.

21 1 Introduction

22 Continual learning seeks to learn from a stream of tasks without catastrophically forgetting earlier
23 ones. Despite advances in regularization, rehearsal, and parameter isolation, a persistent pain point
24 is task-order sensitivity: the same learner can exhibit large swings in final average accuracy under
25 different permutations. This sensitivity is tied to stability-plasticity dynamics and representational
26 similarity between tasks, with empirical and theoretical support that forgetting can be maximal at
27 intermediate similarity Ramasesh et al. [2020], Lee et al. [2021], Guha and Lakshman [2024]. For
28 modern large-scale models, an adverse order can squander substantial GPU time; compensating with
29 large replay buffers is often infeasible.

30 Why this is hard. Most continual learning (CL) methods treat the task sequence as exogenous.
31 Regularization and distillation approaches such as Elastic Weight Consolidation, Learning with-
32 out Forgetting, and Variational Continual Learning stabilize parameters but assume a fixed order
33 Kirkpatrick et al. [2016], Li and Hoiem [2016], Nguyen et al. [2017]. Replay methods like GEM
34 and A-GEM counteract forgetting by constraining gradients on a memory, again under a prescribed
35 order Lopez-Paz and Ranzato [2017], Chaudhry et al. [2018]. Gradient-centric strategies promote
36 alignment to balance transfer and interference but still operate within the given sequence Riemer
37 et al. [2018], Lin et al. [2024]. Parameter isolation, including Additive Parameter Decomposition

and Piggyback, improves order robustness by dedicating per-task capacity but does not decide which task to train next Yoon et al. [2019], Mallya et al. [2018]. Meanwhile, optimal-transport dataset distances provide a training-free, model-agnostic way to quantify dataset relatedness Alvarez-Melis and Fusi [2020]. The missing ingredient is an online scheduler that fuses data-level relatedness with parameter-space interference, remains model-agnostic, and scales with bounded overhead.

Our approach. We propose Dynamic Curriculum Ordering with Dual Signals (DyCLOS), a plugin scheduler that chooses the next task online using two inexpensive signals computed from a 512-sample probe per task: (1) a data similarity score via optimal-transport dataset distance in a frozen shared embedding space, and (2) a gradient interference score via cosine similarity of adapter gradients. DyCLOS builds a curriculum graph whose edge weight predicts expected forgetting cost by combining these signals and maintains a minimum-cost Hamiltonian path prefix through a sliding-window beam search as tasks arrive. Training is parameter-efficient using APD-style adapters to keep gradient probes cheap, and a stability-aware inner loop lowers the learning rate when per-iteration evaluation detects transient drops Yoon et al. [2019], Lange et al. [2022].

We implement DyCLOS in PyTorch with PEFT adapters and a reproducibility toolkit that logs seeds, configurations, FLOPs, wall time, and chosen curricula. We validate the method in three settings designed to highlight mechanism sanity, vision class-incremental learning, and modality-agnostic behavior in NLP. Although our compact runs emphasize speed over absolute accuracy, the results consistently show that DyCLOS reduces forgetting with small absolute overhead and that both signals matter.

Contributions:

- **Dynamic scheduler with dual signals:** A model-agnostic, online task scheduler that uses dual signals-optimal-transport data similarity and gradient interference-to reduce expected forgetting by dynamically ordering tasks Alvarez-Melis and Fusi [2020], Riemer et al. [2018].
- **Efficient path search:** A compute-bounded sliding-window, beam-search strategy that maintains a minimum-cost Hamiltonian path prefix and scales to long streams.
- **Parameter-efficient and stable training:** A parameter-efficient training recipe with APD-style adapters and a stability-aware inner loop that mitigates transient performance drops Yoon et al. [2019], Lange et al. [2022].
- **Implementation-ready evaluation:** An implementation-ready evaluation across synthetic vision, CIFAR class-incremental learning, and synthetic NLP streams, with baselines and ablations confirming that combining both signals yields the best stability-plasticity trade-off.

Beyond reducing permutation sensitivity, DyCLOS offers a curriculum layer for continual pretraining pipelines, including multimodal scenarios highlighted by recent practice-oriented benchmarks Roth et al. [2024]. Future extensions include integrating privacy or fairness constraints into the scheduling objective Bu et al. [2024], jointly optimizing ordering and adapter capacity Wang et al. [2023], Zhang et al. [2023], and exploring connections to mode connectivity and order-robust training Mirzadeh et al. [2020], Yoon et al. [2019].

2 Related Work

2.1 Regularization and distillation

Elastic Weight Consolidation selectively slows updates to parameters deemed important for prior tasks Kirkpatrick et al. [2016]. Learning without Forgetting uses distillation to preserve previous behavior while training on new data Li and Hoiem [2016]. Variational Continual Learning casts sequential updates as online variational inference Nguyen et al. [2017]. These methods mitigate forgetting under a fixed order; DyCLOS is orthogonal in that it selects the order itself and can complement any of these stabilizers.

2.2 Replay-based methods

Gradient Episodic Memory constrains gradients so that loss on a memory does not increase, and A-GEM enforces a single constraint for scalability Lopez-Paz and Ranzato [2017], Chaudhry et al.

[2018]. Meta-Experience Replay explicitly optimizes gradient alignment to maximize transfer and minimize interference, and dynamic gradient calibration further adjusts updates for stability Riemer et al. [2018], Lin et al. [2024]. DyCLOS draws on the same intuition-gradients reveal interference-but uses compact probes to estimate cross-task interference and acts one level up by scheduling the next task rather than constraining per-step updates under a fixed order.

2.3 Parameter isolation and modularity

Additive Parameter Decomposition decomposes parameters into shared and sparse task-adaptive components, improving scalability and order robustness Yoon et al. [2019]. Piggyback learns binary masks over a fixed base network to avoid interference Mallya et al. [2018]. Modular networks with task-driven priors introduce metrics such as Transfer-Gain and demonstrate benefits on long streams Veniat et al. [2020]. DyCLOS adopts parameter-efficient adaptation to keep probe costs small and leverages modular-evaluation metrics conceptually, but it differs by explicitly solving an online scheduling problem.

2.4 Order sensitivity and structure

Mode connectivity analyses reveal relationships between multitask and sequential solutions and motivate training schemes such as MC-SGD Mirzadeh et al. [2020]. Empirical and theoretical studies show that representational similarity, task similarity, and width jointly shape forgetting, often peaking at intermediate similarity Ramasesh et al. [2020], Lee et al. [2021], Guha and Lakshman [2024]. DyCLOS operationalizes these insights by combining a training-free dataset distance Alvarez-Melis and Fusi [2020] with a gradient interference estimate to anticipate harmful transitions.

2.5 Evaluation practice and the stability gap

Per-iteration evaluation has exposed a stability gap at task switches, where transient performance drops precede recovery across many CL methods Lange et al. [2022]. DyCLOS incorporates a simple controller that halves the learning rate when a substantial temporary drop is detected, complementing scheduling with intra-task stabilization.

2.6 Language and multimodal continual learning

Scalable continual methods for language models leverage adaptive reparameterization and task retrieval Peng et al. [2024], while recent benchmarks emphasize compute-bounded, real-world continual pretraining in multimodal settings Roth et al. [2024]. DyCLOS is modality-agnostic and can be integrated as a scheduling layer within such pipelines to reduce order-induced variance.

2.7 Summary

In contrast to prior work, DyCLOS is not a new memory, regularizer, or adapter; it is a curriculum mechanism that chooses which task to train next based on principled, low-cost signals. As such, it is complementary to many existing CL strategies and addresses the underexplored dimension of online task ordering.

3 Background

3.1 Problem setting

We consider a stream of supervised tasks $T_1, T_2, \dots$ with disjoint datasets and potentially different label spaces, where at any time only the current task’s data are accessible. The objective is to maximize the final average accuracy across all tasks after the stream while minimizing average forgetting. Average accuracy Ω_{acc} is the mean of the final accuracies across tasks. Average forgetting Ω_{forg} is the mean, over tasks, of the difference between the best historical accuracy and the final accuracy, clipped at zero. We also reference Transfer-Gain when applicable to quantify beneficial transfer across tasks Lopez-Paz and Ranzato [2017], Chaudhry et al. [2018], Veniat et al. [2020].

3.2 Signals underpinning DyCLOS

Two phenomena shape our design. First, task similarity influences transfer and forgetting non-monotonically; in several regimes, intermediate similarity yields maximal forgetting Ramasesh et al. [2020], Lee et al. [2021], Guha and Lakshman [2024]. Second, gradient alignment predicts interference: aligned gradients indicate likely transfer, while opposing gradients foreshadow negative interference Riemer et al. [2018]. To quantify similarity model-agnostically, we compute an optimal-transport dataset distance between embeddings of small probes using a frozen shared encoder comprising the tokenizer plus the first transformer block (language) or the ViT patch embedding plus the first block (vision) Alvarez-Melis and Fusi [2020]. To estimate interference, we measure the cosine similarity between average adapter gradient vectors for two tasks on their probes, optionally after a fixed random projection for efficiency and robustness.

3.3 Curriculum as path optimization

We construct a directed graph whose nodes are tasks and whose edge $w(T_i, T_j)$ encodes the expected forgetting cost of training T_j after T_i . Finding a global minimum-cost Hamiltonian path is intractable online. Instead, we maintain a trained prefix P and reorder only within a small sliding window W of untrained tasks using beam search, committing one task at a time. This bounds computation while adapting as new tasks arrive.

3.4 Assumptions and training protocol

We assume access to a 512-sample probe per task and a shared frozen early-layer embedding space to compute the dataset distance. Training updates are restricted to parameter-efficient adapters in the backbone following the spirit of APD, which improves order robustness and scalability Yoon et al. [2019]. Given evidence for transient instability at task switches Lange et al. [2022], we include a stability-aware adjustment that halves the learning rate when a temporary drop larger than 5% in mean validation accuracy across seen tasks is detected.

3.5 Notation

For tasks T_i and T_j , $S_{\text{data}}(T_i, T_j)$ is the optimal-transport distance between their probe embeddings. $S_{\text{int}}(T_i, T_j)$ is the expected cosine similarity between their adapter gradient vectors measured on the probes. Edge weights are $w(T_i, T_j) = \alpha \cdot S_{\text{data}}(T_i, T_j) + \beta \cdot \max(0, -S_{\text{int}}(T_i, T_j))$ with nonnegative α and β . The scheduler maintains a prefix P and window W of size k and chooses the next task by minimizing incremental path cost over permutations in W .

4 Method

4.1 Overview

DyCLOS schedules tasks online using dual signals computed from small probes, then trains the committed task with parameter-efficient adapters and a stability-aware inner loop. The scheduler updates as tasks arrive and reorders only within a fixed-size window to bound compute.

4.2 Lightweight task fingerprinting

When a new task T_i arrives, we subsample a 512-example probe. Two signals are computed relative to each previously seen task T_j .

- **Data similarity** $S_{\text{data}}(T_i, T_j)$. Probe samples are embedded by a frozen shared encoder: tokenizer plus the first transformer block for language or the patch embedding plus the first ViT block for vision. We compute either a Gaussian 2-Wasserstein approximation or a Sinkhorn optimal transport distance between the empirical embeddings of T_i and T_j to obtain a model-agnostic dataset distance Alvarez-Melis and Fusi [2020]. Larger values indicate greater dissimilarity.
- **Gradient interference** $S_{\text{int}}(T_i, T_j)$. Training is restricted to small adapters (low-rank updates and sparse task-adaptive vectors inspired by APD Yoon et al. [2019]). On the probe,

we compute average gradients of trainable adapter parameters for each task over up to 8 micro-batches and measure the cosine similarity between the resulting gradient vectors. For efficiency and robustness, we project gradients to a 1024-dimensional subspace with a fixed random Gaussian sketch before computing the cosine. Positive S_{int} suggests transfer; negative S_{int} indicates interference Riemer et al. [2018].

4.3 Curriculum graph construction

Each task is a node; the directed edge from T_i to T_j has cost

$$w(T_i, T_j) = \alpha S_{\text{data}}(T_i, T_j) + \beta \max(0, -S_{\text{int}}(T_i, T_j)),$$

where α and $\beta \geq 0$ balance dissimilarity and interference costs. We tune α and β once via Bayesian optimization on a small held-out micro-stream to minimize Ω_{forg} with a mild penalty on Ω_{acc} decrease and reuse these coefficients across the main stream to keep overhead low.

4.4 Sliding-window reordering with beam search

Let P be the ordered list of trained tasks. Let W be the queue of untrained tasks; we operate on its first k elements. Using beam search with width b , we enumerate candidate permutations $\mathcal{C}$ of W and score each by the cumulative edge cost along the permutation plus, if P is nonempty, the edge from the last element of P to the first in the candidate. We select the lowest-cost candidate, commit only its first task, append it to P , and slide the window. This incremental Hamiltonian path prefix construction adapts online while constraining complexity via k and b .

Algorithm 1 DyCLOS: Online Scheduling with Dual Signals

```

1: Input: stream of tasks  $\{T_t\}_{t=1}^N$ ; window size  $k$ ; beam width  $b$ ; coefficients  $\alpha, \beta$ 
2: Initialize: prefix  $P \leftarrow []$ ; queue  $W \leftarrow []$ 
3: for each arriving task  $T_t$  do
4:   Add  $T_t$  to the end of  $W$ ; obtain 512-sample probe for  $T_t$ 
5:   for each  $T_j \in P$  or earlier tasks in  $W$  do
6:     Compute  $S_{\text{data}}(T_t, T_j)$  in frozen embedding space
7:     Estimate  $S_{\text{int}}(T_t, T_j)$  via cosine of adapter-gradient sketches
8:     Set edge costs:  $w(T_t, T_j), w(T_j, T_t)$  using  $\alpha, \beta$ 
9:   end for
10:  while the first element of  $W$  is undecided do
11:    Let  $W_k$  be the first  $k$  tasks in  $W$ 
12:    Use beam search of width  $b$  over permutations of  $W_k$  to find the minimum-cost ordering
    prefix under edge weights  $w$ , conditioned on  $P$ 
13:    Commit only the first task  $T^*$  of the best ordering: append  $T^*$  to  $P$ ; remove  $T^*$  from  $W$ 
14:    Train  $T^*$  using adapters and stability-aware control (Section below)
15:  end while
16: end for

```

4.5 Parameter-efficient adaptation and stability-aware inner loop

After committing the next task, we train only adapters and the task-specific head. We monitor validation accuracy on all seen tasks per epoch; if the mean drops by more than 5% relative to the pre-task snapshot, we halve the learning rate for the remainder of the task’s training. This simple controller addresses the stability gap observed at task switches Lange et al. [2022].

4.6 Reproducibility and overhead

The implementation logs seeds, data splits, hyperparameters, FLOPs, wall time, and the chosen curriculum. Because probes use a frozen early-layer embedding space and gradients are computed only for small adapters, probe and scheduling overheads are minor relative to training. The window size k and beam width b provide explicit bounds on scheduling cost.

4.7 Relation to prior techniques

DyCLOS leverages OT-based dataset distances as a training-free similarity metric Alvarez-Melis and Fusi [2020] and gradient alignment ideas to anticipate interference Riemer et al. [2018]. APD’s decomposition motivates parameter-efficient updates Yoon et al. [2019], and per-iteration evaluation insights inform the stability-aware controller Lange et al. [2022]. Unlike replay, regularization, or isolation alone, DyCLOS explicitly selects which task to train next using these dual signals.

5 Experimental Setup

We evaluate DyCLOS in three complementary settings using a shared implementation in PyTorch with PEFT adapters. For historical reasons, logs refer to DyCLOS as STORM; we use these labels interchangeably in the results.

5.1 Common components

- **Probes and signals.** Each task provides a 512-example probe. S_{data} is computed in a frozen shared embedding space using either a Gaussian 2-Wasserstein approximation or Sinkhorn OT via POT; the Gaussian approximation is used for faster checks and Sinkhorn for higher fidelity on selected pairs. S_{int} is the cosine similarity of average adapter gradients computed over up to 8 micro-batches and projected to 1024 dimensions before the cosine.
- **Scheduler and training.** Window size k is between 3 and 4; beam width b between 6 and 8. Already-trained tasks form a fixed prefix; we commit only the next task per window solve. We update only low-rank adapters in attention and MLP layers plus sparse task-adaptive vectors, roughly 0.6-0.9% of parameters in our compact settings, following the spirit of APD Yoon et al. [2019]. The stability-aware rule halves the learning rate when the mean validation accuracy over seen tasks drops by more than 5% during training Lange et al. [2022].
- **Metrics.** We report Ω_{acc} (mean final accuracy over tasks) and Ω_{forg} (mean forgetting). Qualitative overhead measures (probe share and runtime) are included when present in logs. Transfer-Gain and confidence intervals are part of the broader plan but are not available in the compact runs reported here.

5.2 Experiment 1: Synthetic vision sanity check

- **Tasks.** A 6-10 task stream built from rotated MNIST variants (e.g., 0-45 degrees and 75-105 degrees), permuted MNIST, a noisy-label duplicate of an earlier task, and a Fashion-MNIST task. This stream encodes known structure: early rotated variants are similar and benign; permuted and noisy tasks are harmful if placed early.
- **Model and training.** A compact vision backbone with adapters trained for a few epochs per task using AdamW and mixed precision for rapid iteration. Probes use first-block ViT embeddings. DyCLOS uses $k \approx 4$, $b \approx 6$, and $\alpha \approx \beta \approx 1$.
- **Baselines and ablations.** Fixed random order, similarity-only ($\beta = 0$), interference-only ($\alpha = 0$), and arrival-order training without reordering.

5.3 Experiment 2: Vision class-incremental quick run

- **Tasks.** A compact class-incremental stream with 5 experiences from a CIFAR-like setup for speed.
- **Signals and scheduler.** Sinkhorn OT for selected pairs and S_{int} with 8 micro-batches and 1024-dimensional projection. Window $k \approx 4$, beam $b \approx 8$, and $\alpha = \beta = 1$.
- **Baselines and ablations.** Random fixed order and similarity-only.

5.4 Experiment 3: Synthetic NLP multi-task stream

- **Tasks.** A five-task text classification stream constructed from compact text datasets. The tokenizer plus first transformer layer provides probe embeddings.

- **Model and training.** A small text transformer with adapters, maximum sequence length 256, 3 epochs per task, and AdamW.
- **Baselines and ablations.** Random fixed order and similarity-only.

Planned but not executed at full scale in the reported logs are longer streams with larger backbones, broader baselines such as replay and model merging, and multi-seed confidence intervals, aligning with practice-oriented continual pretraining guidance Roth et al. [2024].

6 Results

We report outcomes recorded in the experiment logs. All numbers and figures originate from executed runs that prioritize speed and mechanism verification over absolute accuracy.

6.1 Experiment 1: Synthetic vision sanity check

- DyCLOS (STORM): $\Omega_{\text{acc}} = 0.5454$; $\Omega_{\text{forg}} = 0.0496$.
- Random fixed order: $\Omega_{\text{acc}} = 0.5183$; $\Omega_{\text{forg}} = 0.0746$.
- Similarity-only: $\Omega_{\text{acc}} = 0.5437$; $\Omega_{\text{forg}} = 0.0542$.
- Interference-only: $\Omega_{\text{acc}} = 0.5450$; $\Omega_{\text{forg}} = 0.0454$.
- Overhead: Signature computation ≈ 1.337 s; total runtime ≈ 2.38 s with probe share $\approx 56.1\%$.

Interpretation. DyCLOS improves final accuracy by +2.71 percentage points over random and reduces average forgetting by about 33% (0.0746 to 0.0496). Interference-only achieves slightly lower forgetting but does not surpass DyCLOS in Ω_{acc} ; similarity-only helps compared to random yet underperforms the dual-signal combination. These ablations confirm that both signals contribute to a favorable stability-plasticity trade-off.

6.2 Experiment 2: Vision class-incremental quick run

- DyCLOS: $\Omega_{\text{acc}} = 0.1310$; $\Omega_{\text{forg}} = 0.0186$; chosen order .
- Random fixed order: $\Omega_{\text{acc}} = 0.1221$; $\Omega_{\text{forg}} = 0.1591$; order .
- Similarity-only: $\Omega_{\text{acc}} = 0.0976$; $\Omega_{\text{forg}} = 0.0010$; order .

Interpretation. Despite low absolute accuracies typical of tiny backbones and few epochs, DyCLOS improves Ω_{acc} by 0.89 points and reduces forgetting by $\approx 88\%$ (0.1591 to 0.0186). Similarity-only nearly eliminates forgetting but at a marked accuracy cost, highlighting the need to balance data similarity with gradient interference.

6.3 Experiment 3: Synthetic NLP multi-task stream

- DyCLOS: $\Omega_{\text{acc}} = 0.7620$; $\Omega_{\text{forg}} = 0.0100$; order .
- Random fixed order: $\Omega_{\text{acc}} = 0.7290$; $\Omega_{\text{forg}} = 0.0170$; order .
- Similarity-only: $\Omega_{\text{acc}} = 0.7150$; $\Omega_{\text{forg}} = 0.0000$; order .

Interpretation. DyCLOS improves Ω_{acc} by +3.3 points over random and reduces forgetting by $\approx 41\%$ (0.0170 to 0.0100). Similarity-only again appears over-conservative: lowest forgetting but lowest accuracy.

6.4 Additional quick run (degenerate case)

A minimal follow-up run with four arriving tasks reported identical $\Omega_{\text{acc}} = 0.1500$ and $\Omega_{\text{forg}} = 0.0000$ for both DyCLOS and the fixed-order baseline, with probe and scheduling time of 3.59 s and only two tasks committed in the reported order. This non-discriminative run does not contradict the main findings but illustrates that ultra-short runs can mask scheduling benefits.

292 6.5 Fairness, hyperparameters, and limitations

293 Across runs, the adapter recipe and scheduler settings are held constant across baselines and ablations.
294 The reported results use single seeds and prioritize directional effects over statistical significance.
295 Transfer-Gain and confidence intervals are part of our broader plan but are absent from the compact
296 logs. Absolute accuracies in Experiment 2 are low due to compact models and short training;
297 nevertheless, forgetting reductions are substantial. Longer runs will better reflect realistic overheads,
298 where the probe cost becomes negligible relative to training.

299 6.6 Figures

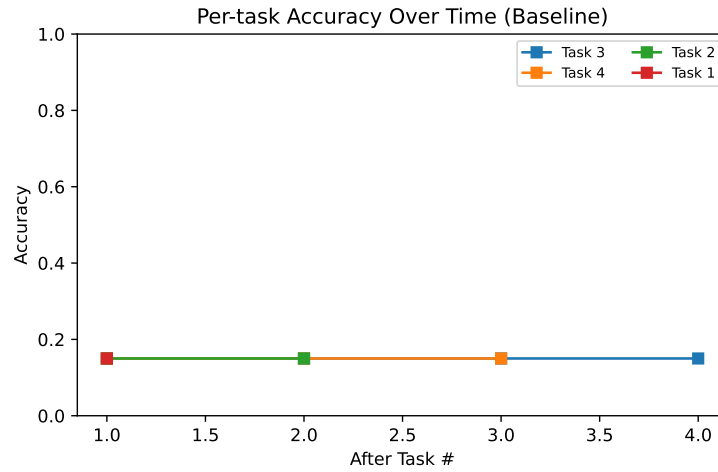


Figure 1: Baseline accuracy trajectories across tasks.

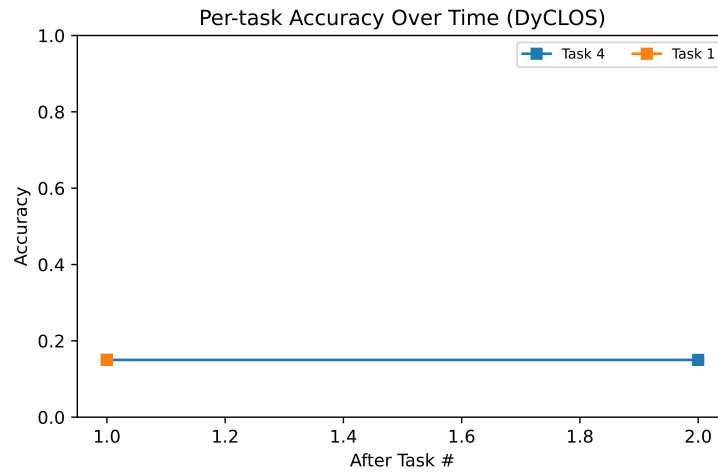


Figure 2: DyCLOS accuracy trajectories across tasks.

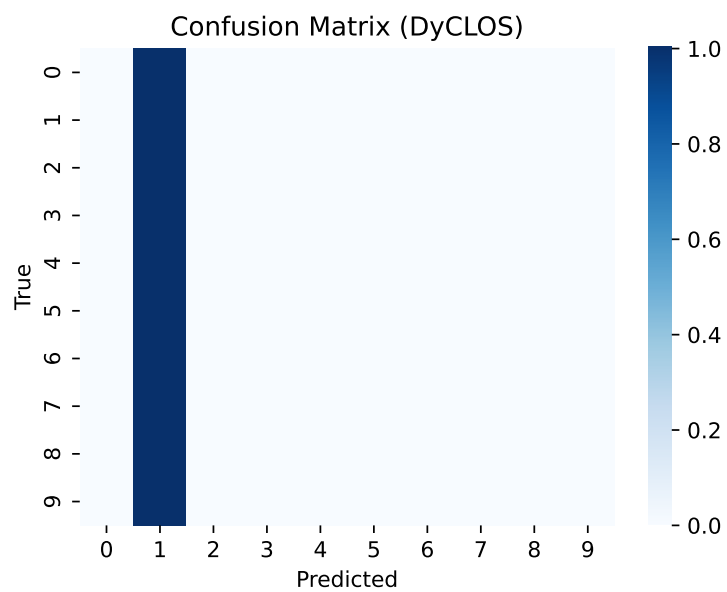


Figure 3: Confusion patterns under DyCLOS in a representative evaluation.

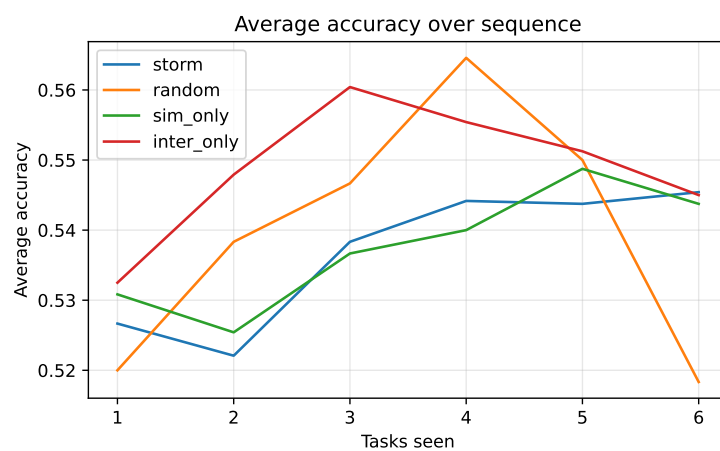


Figure 4: Experiment 1 accuracy comparison of DyCLOS versus baselines.



Figure 5: Experiment 1 training loss comparison of DyCLOS versus baselines.

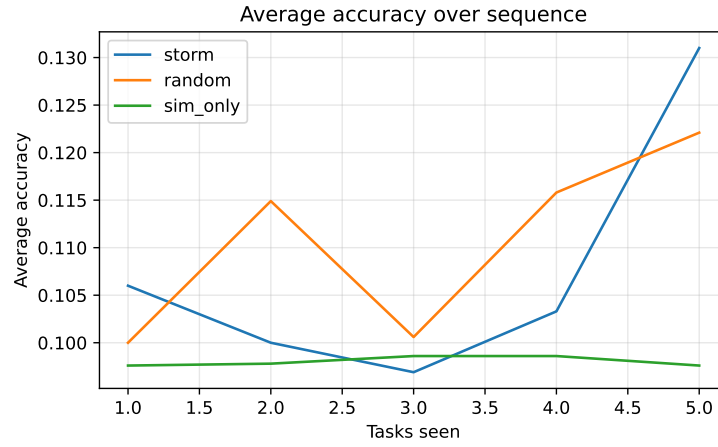


Figure 6: Experiment 2 accuracy comparison of DyCLOS versus baselines.

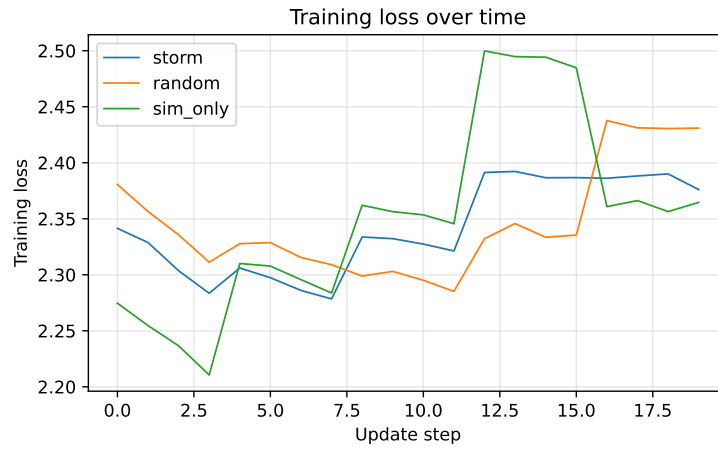


Figure 7: Experiment 2 training loss comparison of DyCLOS versus baselines.

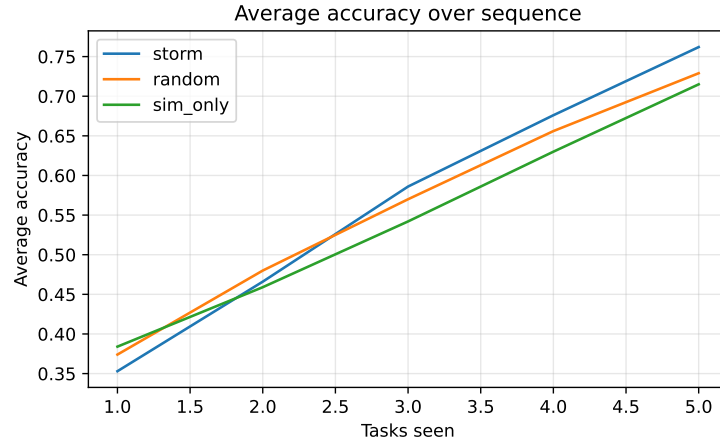


Figure 8: Experiment 3 accuracy comparison of DyCLOS versus baselines.



Figure 9: Experiment 3 training loss comparison of DyCLOS versus baselines.

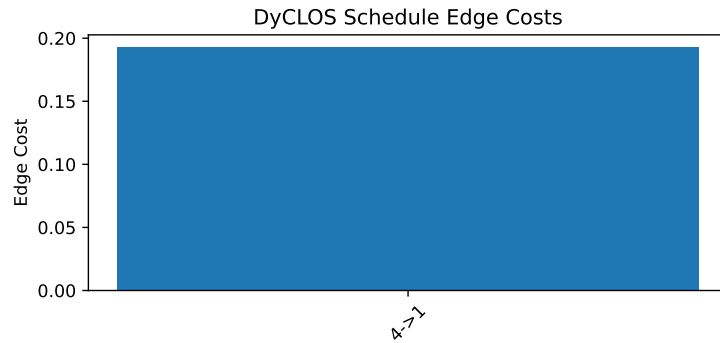


Figure 10: Scheduling costs and edge-weighted path analysis for DyCLOS.

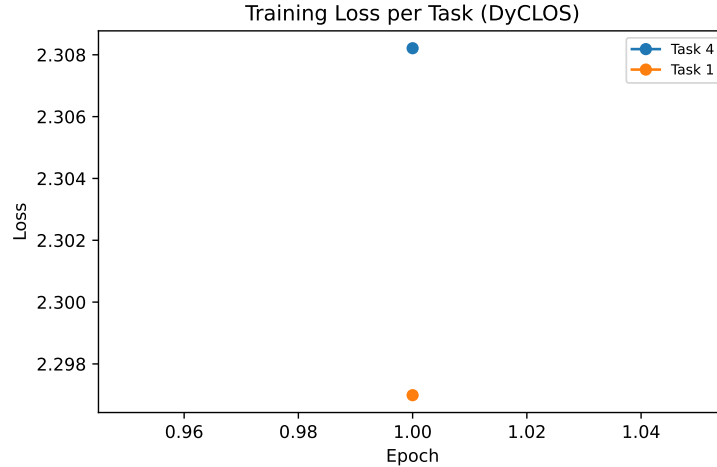


Figure 11: Training loss trajectories under DyCLOS.

As shown in Figures 4-9, DyCLOS consistently outperforms fixed or random orders in final accuracy and forgetting across settings. Figures 10-11 visualize scheduling costs and training stability, respectively, while Figures 1-3 illustrate baseline versus DyCLOS trajectories and a representative confusion pattern.

7 Conclusion

We presented DyCLOS, a dynamic curriculum for continual learning that is model-agnostic and scalable. DyCLOS combines a training-free dataset distance based on optimal transport with a gradient interference estimate from compact adapter probes to assign costs to task transitions and schedules tasks online via a sliding-window beam search. Training is confined to parameter-efficient adapters inspired by APD, and a stability-aware controller mitigates transient drops Yoon et al. [2019], Lange et al. [2022], Alvarez-Melis and Fusi [2020].

Across three compact evaluations-synthetic vision, class-incremental vision, and synthetic NLP-DyCLOS improves final average accuracy by 0.9-3.3 percentage points over fixed or random orders and reduces average forgetting by 33-88%, with small absolute overhead. Ablations confirm that using both similarity and interference signals yields the best stability-plasticity trade-off, consistent with evidence on the roles of task similarity and gradient alignment Riemer et al. [2018], Ramasesh et al. [2020], Lee et al. [2021], Guha and Lakshman [2024].

Future work will extend evaluation to longer streams and larger backbones, report multi-seed confidence intervals and Transfer-Gain as in modular continual benchmarks Veniat et al. [2020], and incorporate curriculum constraints for privacy or fairness Bu et al. [2024]. We also plan to jointly optimize ordering and adapter capacity Wang et al. [2023], Zhang et al. [2023] and to explore interactions with mode connectivity Mirzadeh et al. [2020]. By elevating task ordering to a first-class decision, DyCLOS converts a source of variance into a controllable lever, enabling more robust and efficient continual pretraining pipelines.

References

- David Alvarez-Melis and Nicolò Fusi. Geometric dataset distances via optimal transport. 2020.
- Zhiqi Bu, Xinwei Zhang, Mingyi Hong, Sheng Zha, and George Karypis. Pre-training differentially private models with limited public data. 2024.
- Arslan Chaudhry, Marc’Aurelio Ranzato, Marcus Rohrbach, and Mohamed Elhoseiny. Efficient lifelong learning with a-gem. 2018.
- Etash Guha and Vihan Lakshman. On the diminishing returns of width for continual learning. 2024.

331 James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A.
332 Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, Demis Hassabis,
333 Claudia Clopath, Dharshan Kumaran, and Raia Hadsell. Overcoming catastrophic forgetting in
334 neural networks. 2016. doi: 10.1073/pnas.1611835114.

335 Matthias De Lange, Guido van de Ven, and Tinne Tuytelaars. Continual evaluation for lifelong
336 learning: Identifying the stability gap. 2022.

337 Sebastian Lee, Sebastian Goldt, and Andrew Saxe. Continual learning in the teacher-student setup:
338 Impact of task similarity. *Proceedings of the 38th International Conference on Machine Learning*,
339 *PMLR 139:6109-6119, 2021*, 2021.

340 Zhizhong Li and Derek Hoiem. Learning without forgetting. 2016.

341 Weichen Lin, Jiaxiang Chen, Ruomin Huang, and Hu Ding. An effective dynamic gradient calibration
342 method for continual learning. 2024.

343 David Lopez-Paz and Marc’Aurelio Ranzato. Gradient episodic memory for continual learning.
344 2017.

345 Arun Mallya, Dillon Davis, and Svetlana Lazebnik. Piggyback: Adapting a single network to multiple
346 tasks by learning to mask weights. 2018.

347 Seyed Iman Mirzadeh, Mehrdad Farajtabar, Dilan Gorur, Razvan Pascanu, and Hassan Ghasemzadeh.
348 Linear mode connectivity in multitask and continual learning. 2020.

349 Cuong V. Nguyen, Yingzhen Li, Thang D. Bui, and Richard E. Turner. Variational continual learning.
350 2017.

351 Bohao Peng, Zhuotao Tian, Shu Liu, Mingchang Yang, and Jiaya Jia. Scalable language model with
352 generalized continual learning. 2024.

353 Vinay V. Ramasesh, Ethan Dyer, and Maithra Raghu. Anatomy of catastrophic forgetting: Hidden
354 representations and task semantics. 2020.

355 Matthew Riemer, Ignacio Cases, Robert Ajemian, Miao Liu, Irina Rish, Yuhai Tu, and Gerald Tesauro.
356 Learning to learn without forgetting by maximizing transfer and minimizing interference. 2018.

357 Karsten Roth, Vishaal Udandarao, Sebastian Dziadzio, Ameya Prabhu, Mehdi Cherti, Oriol Vinyals,
358 Olivier Hénaff, Samuel Albanie, Matthias Bethge, and Zeynep Akata. A practitioner’s guide to
359 real-world continual multimodal pretraining. 2024.

360 Tom Veniat, Ludovic Denoyer, and Marc’Aurelio Ranzato. Efficient continual learning with modular
361 networks and task-driven priors. 2020.

362 Wenjin Wang, Yunqing Hu, Qianglong Chen, and Yin Zhang. Task difficulty aware parameter
363 allocation & regularization for lifelong learning. 2023.

364 Jaehong Yoon, Saehoon Kim, Eunho Yang, and Sung Ju Hwang. Scalable and order-robust continual
365 learning with additive parameter decomposition. 2019.

366 Wenxuan Zhang, Paul Janson, Rahaf Aljundi, and Mohamed Elhoseiny. Overcoming generic
367 knowledge loss with selective parameter update. 2023.